



Combining AI Agents with the power of SAP HANA Predictive Analysis Library

Nora von Thenen, SAP
Vitaliy Rudnytskiy, SAP

July, 2026



Meet the speakers



Nora von Thenen

AI Superstar

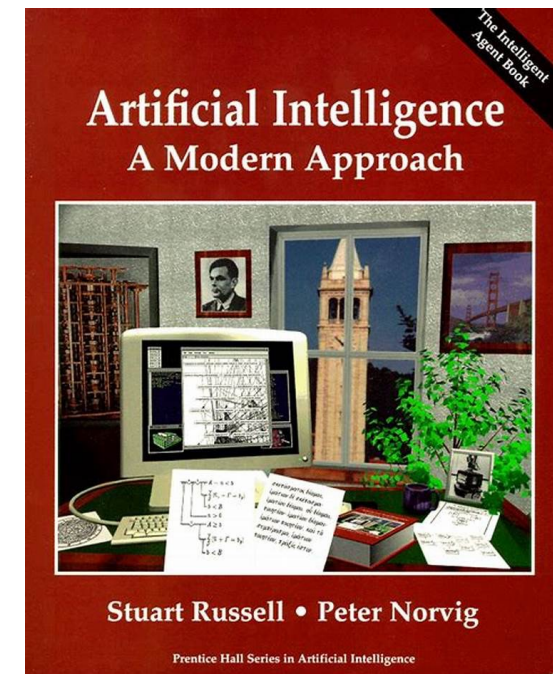
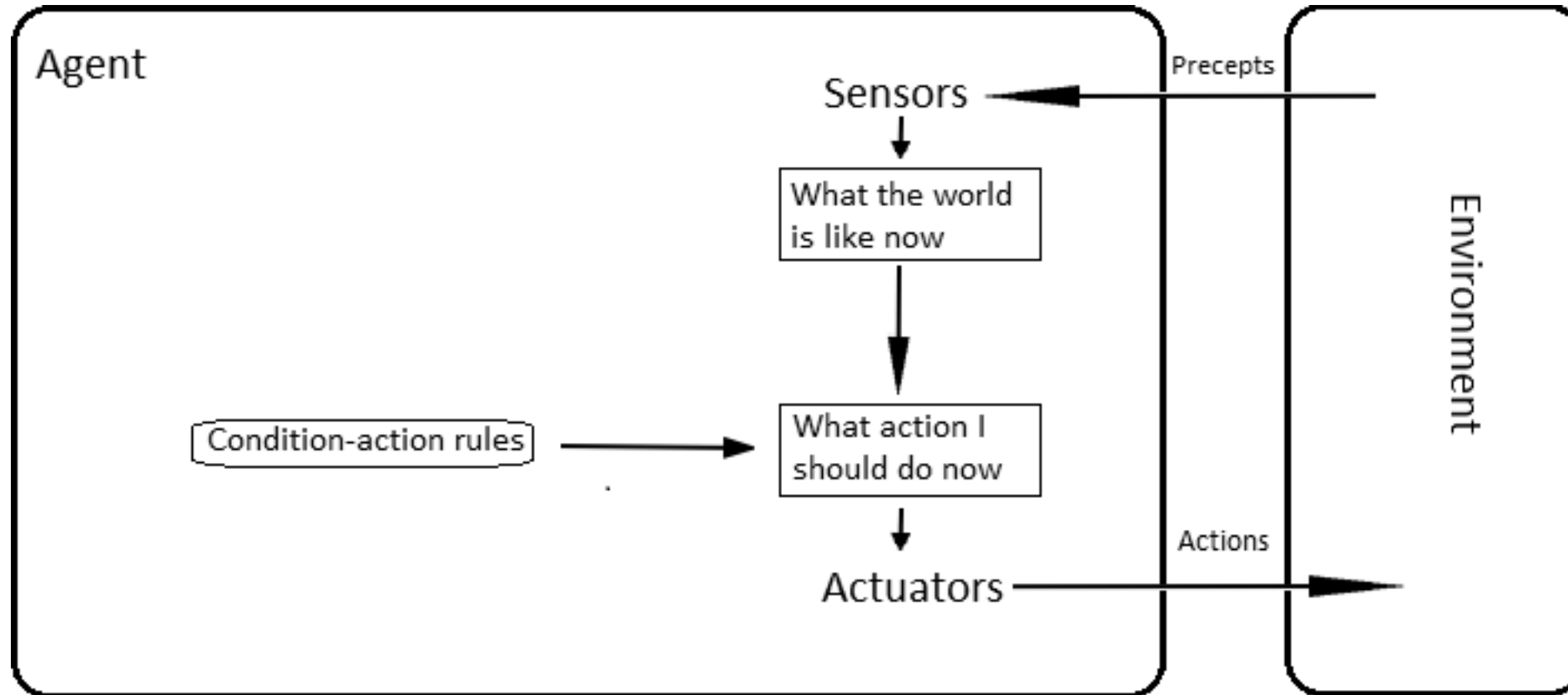


Vitaliy Rudnytskiy

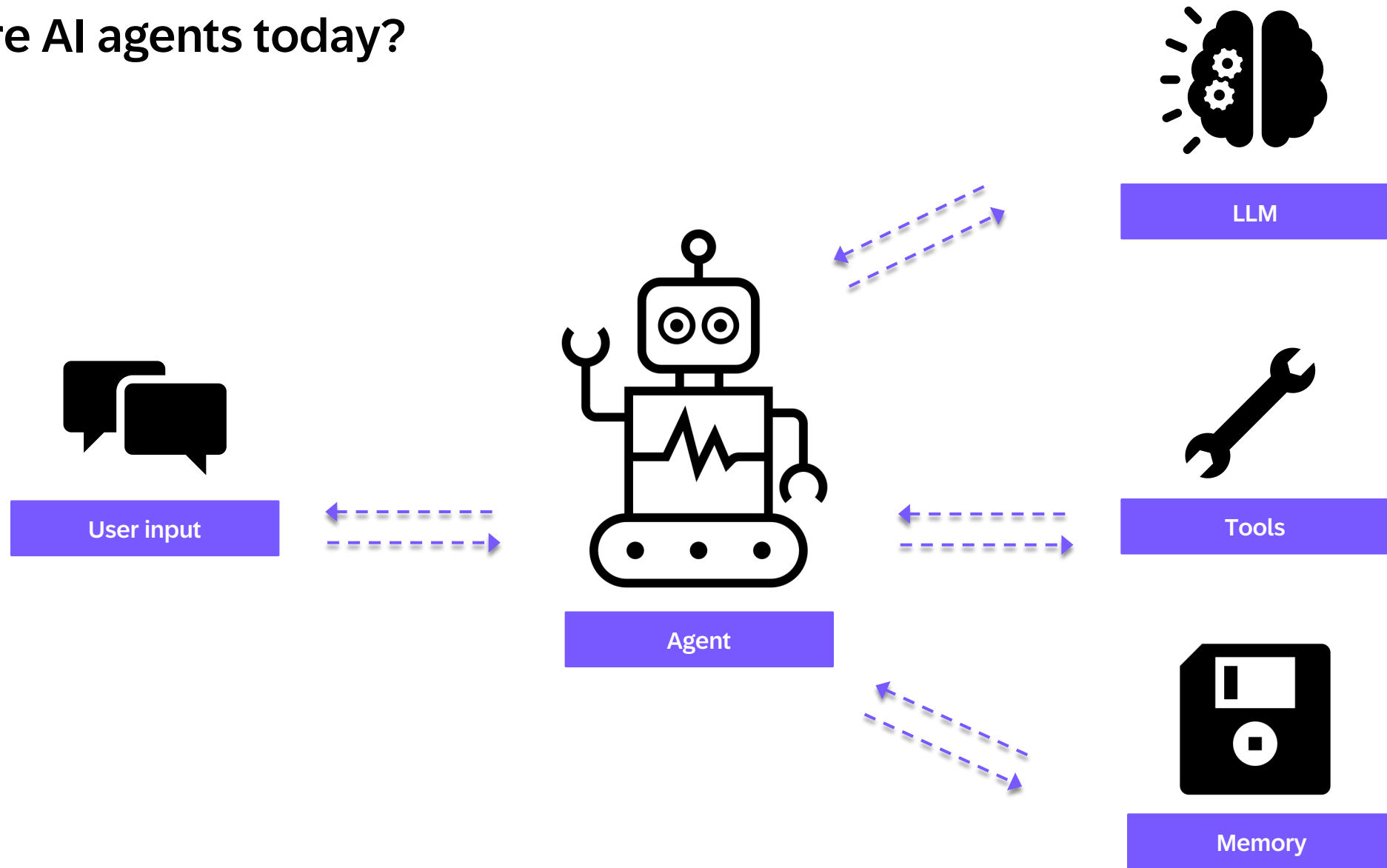
Her driver



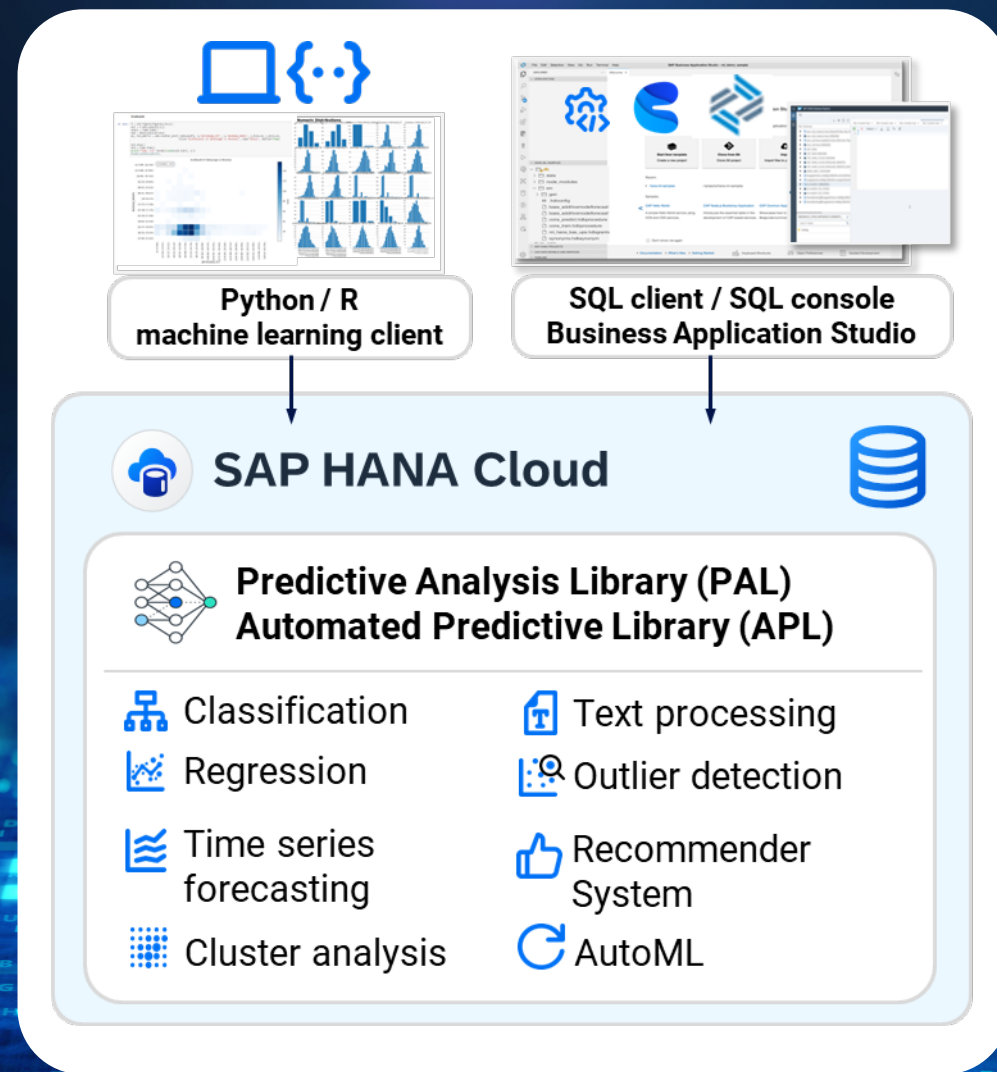
Agent: Anything that perceives its environment and acts upon it



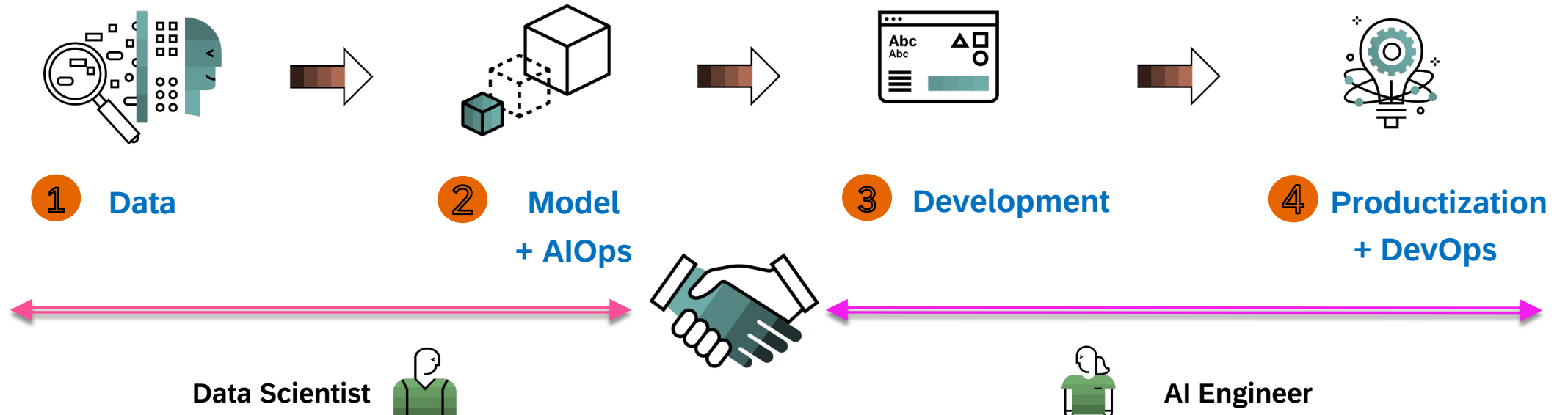
What are AI agents today?



Embedded Machine Learning with SAP HANA Cloud database



Development Approach | AI-based Solutions

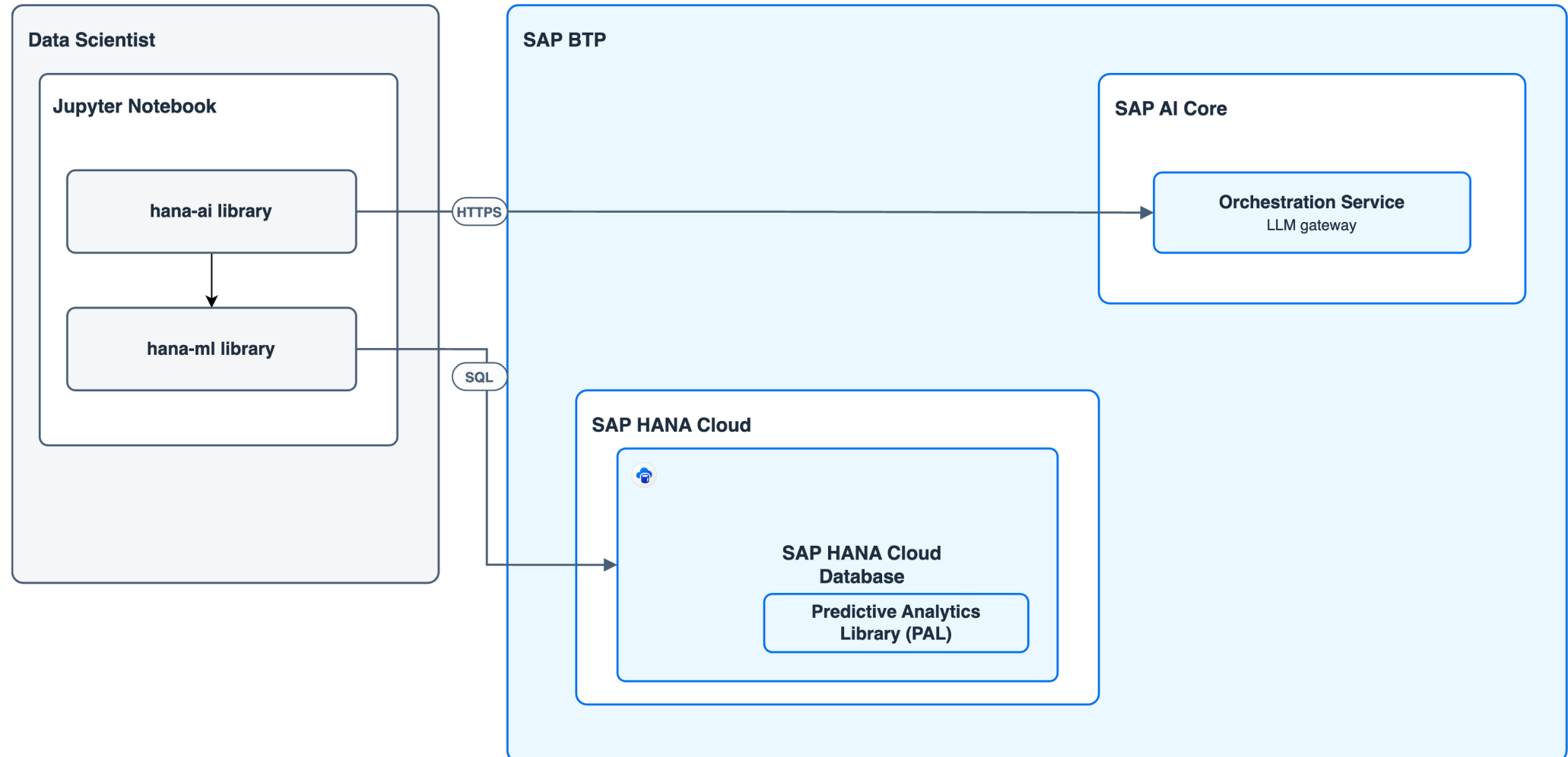




Data Scientist training the ML model

SAP HANA Cloud PAL — hana-ai Python Client

Data scientist runs a local Jupyter Notebook with hana-ai (LLM orchestration) and hana-ml (PAL), connecting to SAP HANA Cloud and SAP AI Core Orchestration



credit: <https://github.com/marianfoo/btp-drawio-skill>



Generative AI Toolkit for SAP HANA Cloud (aka `hana-ai`)

<https://github.com/SAP/generative-ai-toolkit-for-sap-hana-cloud>

hanatechcon26_hanaagents

context_agent_anthropic--claude-4.6-sonnet.ipynb X

context_agent_anthropic--claude-4.6-sonnet.ipynb > M Prepare the tables. We use SALES_REFUNDS dataset and split it into train and test sets.

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outlinevenv (3.13.14) (Python 3.13.14)

```
1 Markdown(chatbot.chat("Now train the table using the suggested model and save as my_hana_ai_model" \
2 "Make the best assumptions to perform the task without asking a user for additional details"))
```

[24] Python

[1/6] Building context
[2/6] LLM reasoning
[3/6] Tool calling
[4/6] Persisting memory
[5/6] Writing structured notes
[6/6] Compaction check

... The model has been successfully trained and saved! Here's a summary:

Detail	Value
Model Name	my_hana_ai_model
Model Version	16
Training Table	SALES_REFUNDS_TRAIN
Key Column	BOOKING_DATE
Target Column	REFUNDS
Algorithm	Automatic Time Series

The Automatic Time Series model has automatically explored and selected the best-fitting pipeline for your `REFUNDS` series.

What would you like to do next?

- **Predict** future values using the test table (`SALES_REFUNDS_PREDICT`)
- **Evaluate** model accuracy against the test set
- **Plot** the forecast results

```
1 Markdown(chatbot.chat("What database tables used to store the `my_hana_ai_model` in?"))
```

[25] Python

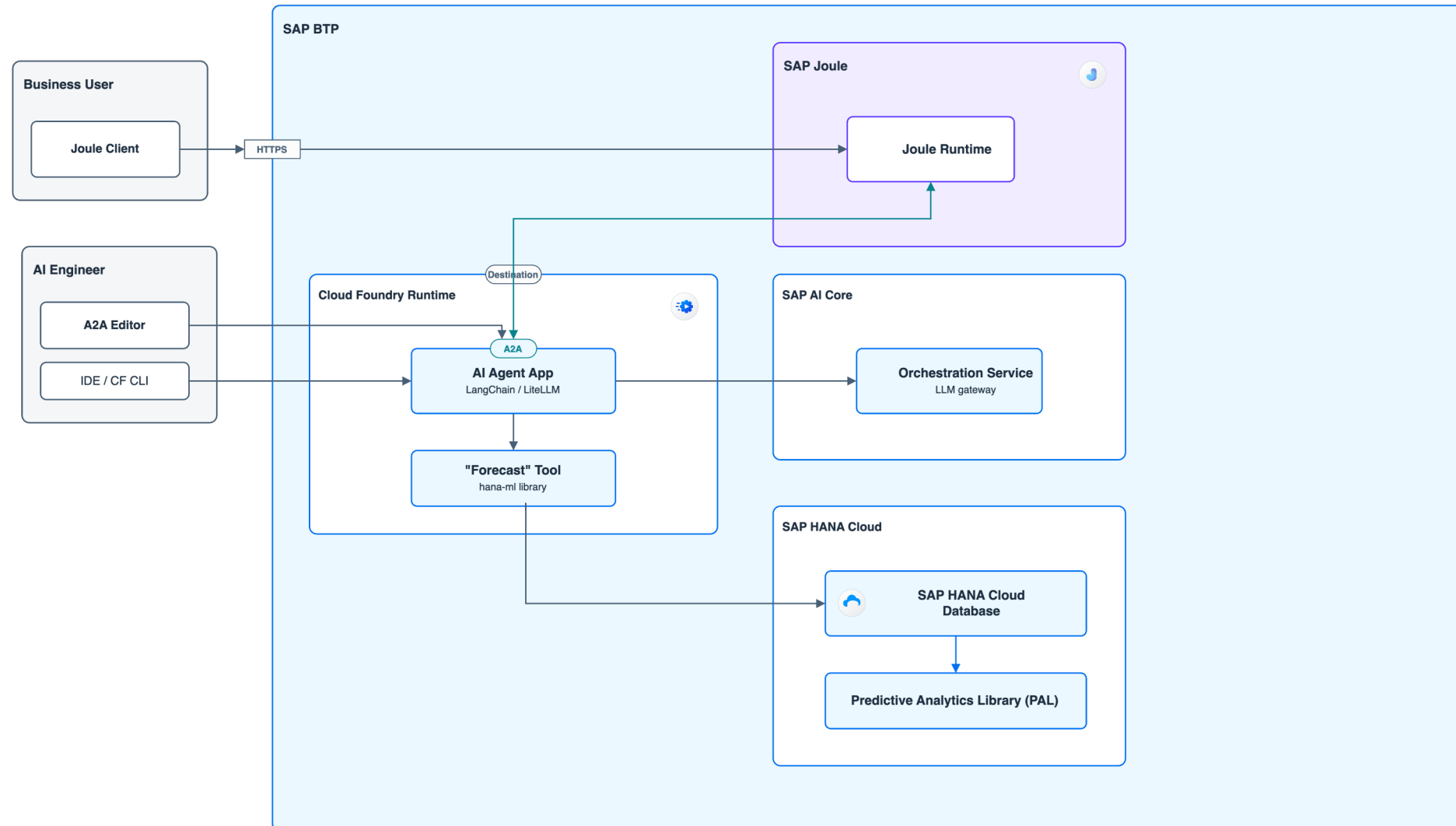
main* 0 0 0 Cell 1 of 37



AI Engineer enabling the ML model from PAL as a tool for an AI agent

Code-Based AI Agent on SAP BTP with PAL

AI Agent deployed on CF Runtime, orchestrated via AI Core, bi-directionally connected to Joule via A2A, and using SAP HANA Cloud PAL via hana-mi



credit: <https://github.com/marianfoo/btp-drawio-skill>

Thank you, Marian, for the

<https://github.com/marianfoo/btp-drawio-skill> 🧐

The screenshot shows a web browser displaying the GitHub repository page for `btp-drawio-skill`. The browser's address bar shows the URL `github.com/marianfoo/btp-drawio-skill/blob/main/README.md`. The repository page header includes the repository name `btp-drawio-skill` and the file name `README.md`. A commit message is visible: `marianfoo and claude docs: add 'Nudge it to a good diagram' guidance to README (#3)`. The page has tabs for `Preview`, `Code`, and `Blame`, with the `Preview` tab selected. The file size is indicated as `261 Lines (179 loc) · 18.8 KB`. The main content area displays the `README.md` file, which includes the title `btp-drawio-skill` and a description: `A Claude Code plugin (and Agent-Skill-compatible standalone skill) that turns a plain-English description into a polished SAP Architecture Center / BTP solution diagram as an editable .drawio file — following the official SAP BTP Solution Diagram Guidelines.` A section titled `Why a dedicated skill?` explains the plugin's purpose. Another section states: `Under the hood it bundles 100 SAP BTP service icons, 71 reference templates, 448 indexed draw.io assets, a validator, and an autofixer — all local, all traceable to SAP's guidelines.` A `Table of contents` section lists links to `Install`, `Recommended models`, `Use it`, `Write better prompts`, `What's bundled`, `How it works`, `Documentation & deep dive`, and `License & credits`.

Model Context Protocol (MCP) at the SAP Architecture Center



A2A and MCP for Interoperability

architecture.learning.sap.com/docs/ref-arch/76ec36#model-context-protocol-mcp

SAP

Architecture Center

SAP Viewpoints

Navigate

Search

K

Filters

114 unique documents found

AI & Machine Learning (37)

SAP Document AI

Generative AI on SAP BTP

Transforming Enterprise Data Strategy with SAP Business Data Cloud

Identity Access Management

Integrating and Extending Joule

Embodied AI Agents & Robotics

Agentic AI & AI Agents

A2A and MCP for Interoperability

Low-Code AI Agents with Joule Studio

Pro-Code AI Agents on SAP BTP

Integrating AI Agents with Joule

Integrating Joule Agents into Your Ecosystem

Agents for Structured Data

Model Context Protocol (MCP)

Model Context Protocol (MCP) is an open standard that defines how AI models and agents can discover, understand and interact with external tools and their surrounding context. It acts as a universal adapter, allowing agents to consume tools—from simple functions to complex APIs—without needing to know their underlying implementation details.

Key Functions of MCP:

- Tool Discovery:** MCP servers expose a manifest that describes the available tools, their functions and their input/output schemas. This allows an agent to dynamically discover what actions it can perform.
- Standardized Interaction:** It provides a consistent way for an agent to call a tool and receive a response, abstracting away the specifics of the tool's implementation (e.g., REST API, OData service, or a simple function).
- Decoupling:** By placing tools behind an MCP interface, they can be developed, versioned and deployed independently of the agents that use them. This is critical for maintainability and scalability.

At SAP, MCP is used to provide Joule Agents with semantically enriched access to SAP business capabilities and domain knowledge, including content from SAP Knowledge Graph. For external interoperability between vendors and third-party agents, SAP uses the Agent2Agent (A2A) protocol as the preferred approach, ensuring enterprise-grade security, governance and controlled access.

Architecture

Agent2Agent (A2A) Protocol

Model Context Protocol (MCP)

Agent Gateway (Inbound)

MCP Gateway in Integration Suite

Bring Your Own Agent (Outbound)

Simplified Flow

SAP's Commitment to Open Standards

Examples

PUBLIC

12

Agent2Agent Protocol (A2A) at the SAP Architecture Center



Architecture Center

SAP Viewpoints

Navigate

Search

K

Filters

114 unique documents found

AI & Machine Learning (37)

SAP Document AI

Generative AI on SAP BTP

Transforming Enterprise Data Strategy with SAP Business Data Cloud

Identity Access Management

Integrating and Extending Joule

Embodied AI Agents & Robotics

Agentic AI & AI Agents

A2A and MCP for Interoperability

Low-Code AI Agents with Joule Studio

Pro-Code AI Agents on SAP BTP

Integrating AI Agents with Joule

Integrating Joule Agents into Your Ecosystem

Agents for Structured Data

Agent2Agent (A2A) Protocol

The **Agent2Agent (A2A) protocol** is an open standard for communication and collaboration between autonomous AI agents. It enables one agent to delegate tasks to another, inquire about its capabilities and exchange information in a structured way.

Key Functions of A2A:

Agent Integration:

A2A provides the standard contract for integrating external developed agents (e.g., pro-code agents) into the SAP ecosystem. By exposing an agent as an A2A server, it becomes discoverable and usable by other systems, notably Joule.

Task Delegation:

It allows for the creation of sophisticated, multi-agent workflows where a primary agent can orchestrate specialized sub-agents to solve a complex problem.

Interoperability:

Because A2A is an open standard, agents built with different frameworks and by different teams or organizations can communicate seamlessly, fostering a diverse and powerful agent landscape.

Model Context Protocol (MCP)

For pro-code agents, exposing an **A2A-Server endpoint** is the primary mechanism for integrating with Joule. Joule acts as an A2A client, sending requests to the agent and processing its responses according to the A2A-defined contract.

Architecture

Agent2Agent (A2A) Protocol

Model Context Protocol (MCP)

A2A EditorDocumentationPlayground

Getting Started

A2A Editor provides React components for editing and testing A2A (Agent-to-Agent) Protocol Agent Cards.

Installation

npm install @open-resource-discovery/a2a-editor

Quick Start

Full PlaygroundEditorRead-Only Card

Agents

Solar System Explorer

Code Review Assistant

Recipe Chef

Mocked LLM

weather

planets

fun

```
1 {
2   "name": "Echo Agent",
3   "url": "mock://echo",
4   "version": "1.0.0",
5   "description": "A simple echo agent that returns exactly what you send. Useful for testing connectivity and message format.",
6   "provider": {
7     "name": "Agent Card Editor",
8     "organization": "Mock"
9   },
10  "capabilities": {
11    "streaming": false,
12    "pushNotifications": false
13  },
14  "skills": [
15    {
16      "id": "echo",
17      "name": "Echo",
18      "description": "Echoes back the user message verbatim.",
19      "tags": [
20        "test",
21        "echo",
22        "debug"
23      ],
24      "function": () => {
25        return {
26          "text": "Echoed: " + message,
27          "format": "text/plain"
28        };
29      }
30    }
31  ]
32}
```

Echo Agent v1.0.0

Protocol: 0.3

A simple echo agent that returns exactly what you send. Useful for testing connectivity and message format.

Agent Card EditorMock

mock://echo

Connection

Connected

Agent URL: mock://echo

Disconnect

No Authentication

Input/Output Modes

Input: text/plain

Output: text/plain

The complete playground with Monaco editor and chat. Try the mock agents to test the interface.

import { AgentPlayground } from "@open-resource-discovery/a2a-editor";

Installation

Quick Start

Available Components

Peer Dependencies

Features

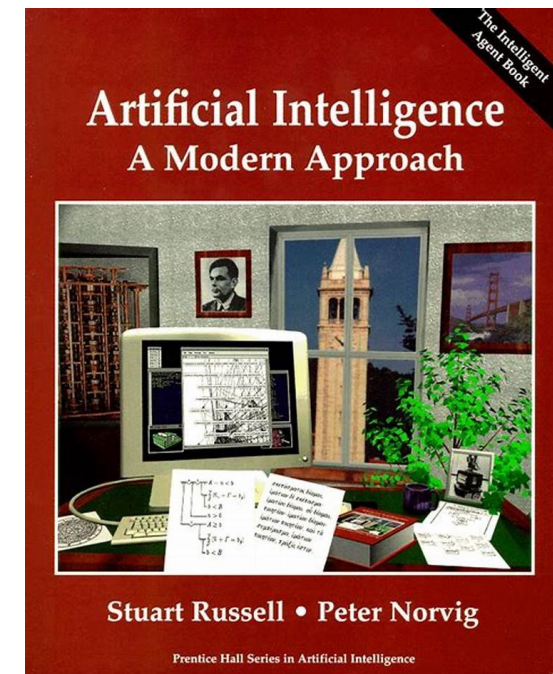
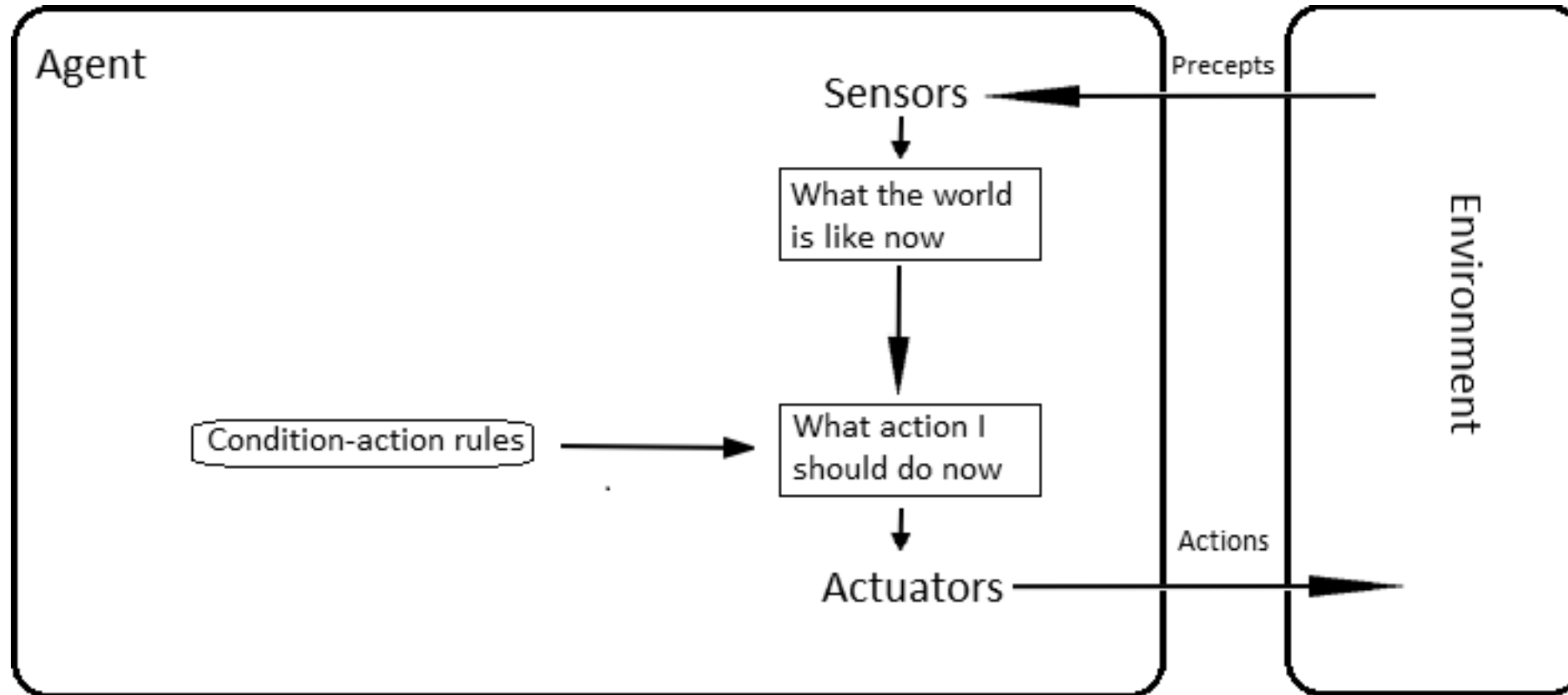
Docker

Funded by the European Union

Supported by: Federal Ministry for Economic Affairs and Energy

on the basis of a decision by the German Bundestag

Agent: Anything that perceives its environment and acts upon it



Origins of the current “Agentic AI” hype

DeepLearning.AI

THE BATCH

Explore CoursesAI NewsletterCommunityMembershipStart Learning

Weekly IssuesAndrew's LettersData PointsML ResearchBusinessScienceCultureHardwareAICareersAboutSubscribe

The Batch > Letters > Article

Welcoming Diverse Approaches Keeps Machine Learning Strong

What technology counts as an “agent”? Instead of arguing, let's consider a spectrum along which various technologies are “agentic.”

LettersTechnical Insights

PublishedJun 12, 2024

Reading time2 min read

IT'S AN AGENT!

NO, IT'S NOT!

IT'S AGENTIC!

Share

Xfili

Dear friends,

One reason for machine learning's success is that our field welcomes a wide range of work. I can't think of even one example where someone developed what they called a machine learning algorithm and senior members of our community criticized it saying, “that's not machine learning!” Indeed, linear regression using a least-squares cost function was used by mathematicians Legendre and Gauss in the early 1800s — long before the invention of computers — yet machine learning has embraced these algorithms, and we routinely call them “machine learning” in introductory courses!

Line chart showing the trend of 'agentic AI' over time. The x-axis represents time from May 2, 2022, to Feb 8, 2026. The y-axis represents a value from 0 to 50. The chart shows a sharp increase starting around late 2023, peaking in early 2024, and then fluctuating at a high level. A tooltip for the period Jun 30 - Jul 6, 2024 shows a value of 1.

Time Period	Value
Jun 30 - Jul 6, 2024	1

PUBLIC

AI Agents... or “Digital Minions”?

QUT Business School

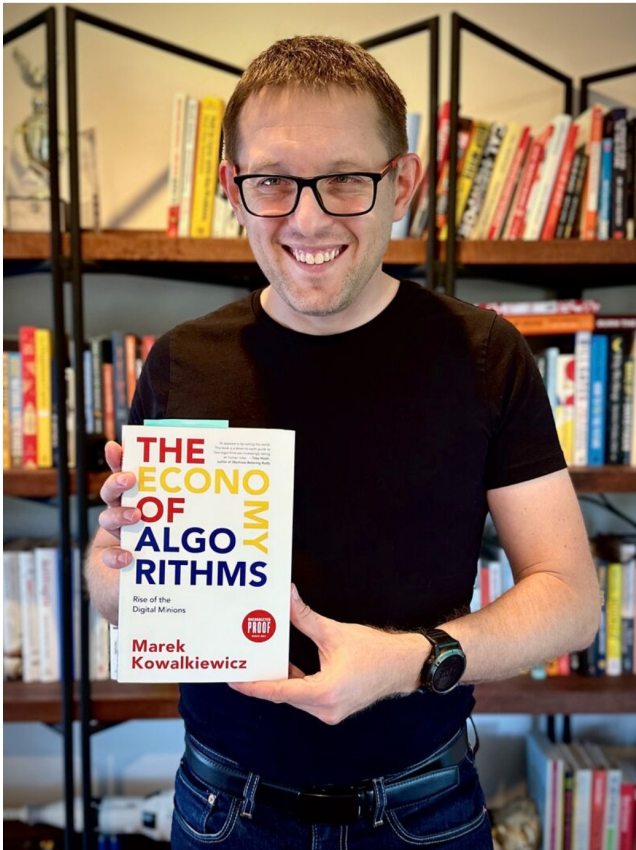
9,970 followers
6mo · 🌐

+ Follow ...

Congratulations to [Prof. Marek Kowalkiewicz](#) for winning the Technology category of this year's Australian Business Book Awards with his book, *The Economy of Algorithms*.

Curious to explore this award-winning work? Grab your copy here: <https://lnkd.in/gTM3RMmt>

#QUTBusiness



📷 Prof Marek Kowalkiewicz, a Brisbane-based lecturer in AI has released a book *The Economy of Algorithms: AI and the Rise of the Digital Minions*. Photograph: Dan Peled/The Guardian



🕒 This article is more than 1 year old

'I welcome our digital minions': the Silicon Valley insider warning about algorithms - while embracing them

Marek Kowalkiewicz has made a career of harnessing new technology. But on the question of if algorithms have agency, he realised he was wrong

By [Joe Hinchliffe](#)



Danke schön //
Дякуємо //
Dziękujemy!

SAP Developer Advocates

Nora von Thene
Vitaliy Rudnyiyskiy

